



Sec: XI_STU_IC_IIT

Time: 3 Hrs.

WARMUP FTM-02

Date : 17-06-2024

Max. Marks : 180

JEE MAIN 2022 Model

PHYSICS

Section	Question Type	+Ve Marks	-Ve Marks	No.of Qs.	Total marks
Sec – I (Q.N : 1 – 10)	Questions with Single Correct Choice	4	–1	10	40
Sec–II (Q.N : 11 – 15)	Questions with Numerical Answer Type(+ / –Number)	4	–1	5	20
Total				15	60

CHEMISTRY

Section	Question Type	+Ve Marks	-Ve Marks	No.of Qs.	Total marks
Sec – I (Q.N : 31 – 50)	Questions with Single Correct Choice	4	–1	10	40
Sec–II (Q.N : 51 – 60)	Questions with Numerical Answer Type (+ / – Number)	4	–1	5	20
Total				15	60

MATHEMATICS

Section	Question Type	+Ve Marks	-Ve Marks	No.of Qs.	Total marks
Sec–I (Q.N : 61 – 80)	Questions with Single Correct Choice	4	–1	10	40
Sec–II (Q.N : 81 – 90)	Questions with Numerical Answer Type (+ / – Number)	4	–1	5	20
Total				15	60

Exam Syllabus

PHYSICS	40% : KINEMATICS: Frame of Reference, Mathematical Expression and definition of position, distance, and displacement Avg velocity & instantaneous velocity, Avg speed & instantaneous speed, acceleration 40% : Unit and Dimension: Significant figures, Errors in measurements Problems discussion from material. 20% : MATHEMATICAL PHYSICS: Application of differentiation in physics, Maxima & Minima Integration and its application in physics. Unit and Dimension: Introduction of physics, physical quantities, and physical laws Fundamental and derived quantities and ways to write dimension, Unit and Dimension Uses of dimensional analysis (Principle of homogeneity).
CHEMISTRY	40% : Question practice on concentration Concept of limiting reagent: Use of stoichiometry in balanced equation and limiting reagent concept. 40% : Concentration terms: Molarity, molality, mole fraction and their interconversions, relation between density and molarity. Concentration terms and their interconversions, relation between density and molarity 20% : Nature of Matter, Properties of matter and their measurement SI unit, Mass weight, Volume temperature, need for standard reference, Scientific notation, Precision and accuracy, Significant figure and calculation involving significant figure. Law of chemical combination: (i) Law of conservation of mass, (ii) Law of constant combination, (iii) Law of multiple proportion, (v) Gay Lussac's law, (vi) Avagadro law, (vii) Dalton's atomic theory atomic mass Mole concept: Calculation based on mole concept, concept of gram atom, gram molecule Percentage by mass (w/w), percentage by volume (v/v).
MATHEMATICS	40%: TRIGONOMETRIC FUNCTIONS Angle, system of measurement of angles, trigonometric function, Values of trigonometric function for some specific angles Trigonometric ratios of allied angles, transforming product into sum or difference, Transforming the sum or difference into product. 40%: Introduction to HP, HM, Relation between AM, GM, and HM Difference method, Vn method, special series 20%: SEQUENCE AND SERIES Introduction of Sequence G.P., sumies, nth term of AP. Properties of A.P., Sum of n term of an AP, Arithmetic Mean nth term of G.P., sum of n term of GP. Properties of G.P., Geometric mean, A.G.P.

Rough Work

PHYSICS

Max.Marks:60

SECTION - I

(SINGLE CORRECT ANSWER TYPE)

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- The acceleration due to gravity (g) is measured on earth's surface by using a simple pendulum. If time period of the pendulum $T = (4.00 \pm 0.01)\text{s}$ and length of pendulum $l = (4.00 \pm 0.01)\text{m}$ were measured, then value of g upto correct significant figures, is (use $\pi^2 = 10.0$)
 (A) $(9.8 \pm 0.2)\text{m/s}^2$ (B) $(10.0 \pm 0.1)\text{m/s}^2$
 (C) $(9.8 \pm 0.8)\text{m/s}^2$ (D) $(10.0 \pm 0.8)\text{m/s}^2$
- The position (x) of a particle at time t is given by the relation $x = \frac{A}{B}(1 - e^{-Ct})$, where A, B and C are constants and $C > 0$. The dimension of $\frac{AC}{B}$ is.
 (A) $[M^0 L^2 T^{-1}]$ (B) $[M^0 L T^{-1}]$
 (C) $[M^0 L^{-1} T]$ (D) $[M^0 L T]$
- The pair of physical quantities having different dimensions is
 (A) Plane angle and solid angle
 (B) Angular velocity and frequency
 (C) Relative velocity and relative density
 (D) Impulse and linear momentum
- The position (x) of a particle varying with time (t) is given by $x = (t^2 - 6t)\text{m}$. The distance covered by particle in first 6 s will be
 (A) 6 m (B) 9 m (C) 15 m (D) 18 m
- A vehicle travels half of the distance with speed v_1 and other half of distance in two equal time intervals with speed v_2 and v_3 . The average speed of vehicle is
 (A) $\frac{2v_1v_2+2v_1v_3}{v_1+v_2+v_3}$ (B) $\frac{2v_1v_2+2v_1v_3}{2v_1+v_2+v_3}$ (C) $\frac{3v_1v_2v_3}{v_1v_2+v_2v_3+v_1v_3}$ (D) $\frac{v_1v_2+v_1v_3}{2v_1+v_2+v_3}$
- The result of expression $\frac{0.301 \times 0.27}{1.149}$ in appropriate significant figures is
 (A) 0.07 (B) 0.071 (C) 0.07073 (D) 0.0707
- A planet moves around the sun in a circular orbit. If the period of revolution of the planet T depends upon the radius R of its orbit, the mass M of the sun and the universal gravitational constant G . The dimensional formula of time period is
 (A) $[M^{-\frac{1}{2}}G^{-\frac{1}{2}}R^{\frac{3}{2}}]$ (B) $[M^{-1}G^{-1}R^3]$
 (C) $[M^{-1/2}G^{1/2}R^{-3/2}]$ (D) $[M^{1/2}G^{1/2}R^3]$
- The value of 10 J in a system of units, which has 10 cm, 1 kg and 30 s as the units of length, mass, and time respectively, is
 (A) 2×10^4 units (B) 9×10^5 units
 (C) 15×10^5 units (D) 18×10^5 units

Rough Work

9. The equation of state for 1 mole of a real gas is given by $\left(P + \frac{a}{V^2}\right)(V - b) = RT$. The dimension of PV is same as that of (where P, V and T are pressure, volume and temperature respectively)
 (A) Force (B) Surface tension (C) Pressure (D) Work
10. For a particle moving along a straight line, its position x is given by $x = 2e^{-2t}$, where t is time. Relation between its velocity ' v ' and acceleration ' a ' is given by
 (A) $a = -v$ (B) $a = v$ (C) $a = -2v$ (D) $a = 2v$

SECTION-II

(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only. Have to Answer any 5 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

11. The order of magnitude of 2^{20} will be _____.
12. The area of a blot of ink is growing such that after t second, its area is given by $A = (3t^2 + 7)\text{cm}^2$. Calculate the rate of increase of area in cm^2/sec at $t = 5$ seconds.
13. A particle starts from rest and traverses a distance l with uniform acceleration, then moves uniformly over a further distance $2l$ and finally comes to rest after moving a further distance $3l$ under uniform retardation. Assuming the entire motion to be rectilinear motion, the ratio of average speed over the journey to the maximum speed on its way is $x/10$. Find the value of x _____.
14. The specific resistance σ of a thin wire of radius r cm, resistance $R\Omega$ and length L cm is given by
- $$\sigma = \frac{\pi r^2 R}{L}$$
- If $r = 0.26 \pm 0.02$ cm, $R = 32 \pm 1\Omega$ and $L = 78 \pm 0.01$ cm, The percentage error in σ is. ____.
15. A gas bubble, from an explosion under water, oscillates with a period T proportional to $p^a d^b E^c$, where p is the pressure, d is the density of water and E is the total energy of the explosion. Find the value of $\frac{6a^2}{b+c}$.

Rough Work

CHEMISTRY**Max.Marks:100****SECTION – I****(SINGLE CORRECT ANSWER TYPE)**

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

16. A sample of CaCO_3 has $\text{Ca} = 40\%$, $\text{C} = 12\%$ and $\text{O} = 48\%$. If the law of constant proportions is true, then the mass of Ca in 5g of CaCO_3 from another source will be:
 A) 2.0 g B) 0.2 g C) 0.02 g D) 20.0g
17. Number of atoms in 4.25g of NH_3 is:
 A) 6.023×10^{23} B) $4 \times 6.023 \times 10^{23}$ C) 1.7×10^{24} D) $4.5 \times 6.023 \times 10^{23}$
18. In the reaction, $\text{N}_2 + 3\text{H}_2 \longrightarrow 2\text{NH}_3$, the ratio of volumes of nitrogen, hydrogen and ammonia is 1: 3: 2. These figures illustrate the law of:
 A) Constant proportions B) Gay-Lussac
 C) Multiple proportions D) Reciprocal proportions
19. 1 amu is equal to:
 A) 1.00758g B) 0.000549g C) 1.66×10^{-24} g D) 6.02×10^{-23} g
20. Which has the maximum number of atoms?
 (A) 24g C(12) B) 56g Fe(56) C) 27g Al(27) D) 108g Ag (108)
21. An element, X has the following isotopic composition,
 $^{200}\text{X} : 90\%$
 $^{199}\text{X} : 8\%$
 $^{202}\text{X} : 2\%$
 The weighted average atomic mass of the naturally occurring element 'X' closest to:
 A) 201 amu B) 202 amu C) 199 amu D) 200 amu
22. The molarity of a solution containing 50g of NaCl in 500g of a solution and having a density of $0.936\text{g}/\text{cm}^3$ is:
 A) 1.5 M B) 1.6 M C) 1.8 M D) 1.2 M
23. 20 mL of 0.5 M HCl is mixed with 30 mL of 0.3 M HCl, the molarity of the resulting solution is:
 A) 0.8 M B) 0.53 M C) 0.38 M D) 0.83 M
24. 3 gm of Mg is burnt in a closed vessel containing 3 gm of oxygen. The weight of excess reactant left is -
 A) 0.5 gm of oxygen B) 1.0 gm of oxygen C) 1.0 gm of Mg D) 0.5 gm of Mg
25. Find out the molarity if 0.4g NaOH is present in 400 ml of solution.
 A) 0.025 B) 0.25 C) 2.5 D) 25

Rough Work

SECTION-II

(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only. Have to Answer 5 questions** and questions will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

26. Calculate the mass of 1.5-gram molecule of sulphuric acid.
27. Calculate the volume at NTP occupied by 6.25g of nitrogen. (Take molar volume = 22.4L)
28. 7.45g of potassium chloride is dissolved in 100g of water. What will be the molality of the solution? (KCl = 74.5 amu)
29. What is the normality of 1M H_3PO_4 solution?
30. 6 moles of Na_2CO_3 are reacted with 4 moles of HCl solution. Find out no. of moles of CO_2 formed?

MATHEMATICS

Max.Marks:100

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

31. If $\cos x + \cos^2 x = 1$, then the value of $\sin^{12} x + 3\sin^{10} x + 3\sin^8 x + \sin^6 x - 1$, is equal to
A) 2 B) 1 C) -1 D) 0
32. The value of $\sin \frac{\pi}{16} \sin \frac{3\pi}{16} \sin \frac{5\pi}{16} \sin \frac{7\pi}{16}$ is
A) $\frac{\sqrt{2}}{16}$ B) $\frac{1}{8}$ C) $\frac{1}{16}$ D) $\frac{\sqrt{2}}{32}$
33. Let $x + y + z = 15$ if 9, x, y, z, a are in A.P.; while $[1/x] + [1/y] + [1/z] = 5/3$ if 9, x, y, z, a are in H.P., then what will be the value of a?
A) 1 B) 2 C) 1/2 D) 3
34. If $\sqrt{\frac{1+\cos A}{1-\cos A}} = \frac{x}{y}$, then the value of $\tan A$ is equal to
A) $\frac{x^2 + y^2}{x^2 - y^2}$ B) $\frac{2xy}{x^2 + y^2}$ C) $\frac{2xy}{x^2 - y^2}$ D) $\frac{2xy}{y^2 - x^2}$
35. The value of $\cos^4 \theta - \sin^4 \theta$ is equal to
A) $1 + 2\sin^2\left(\frac{\theta}{2}\right)$ B) $2\cos^2 \theta - 1$ C) $1 - 2\sin^2\left(\frac{\theta}{2}\right)$ D) $1 + 2\cos^2 \theta$

Rough Work

36. If $S_n = nP + \frac{1}{2}n(n-1)Q$, where S_n denotes the sum of the first n terms of an AP, then the common difference is
 A) $P + Q$ B) $2P + 3Q$ C) $2Q$ D) Q
37. If the $(p + q)^{\text{th}}$ term of a geometric series is m and the $(p - q)^{\text{th}}$ term is n , then the p^{th} term is
 A) $(mn)^{1/2}$ B) mn C) $m + n$ D) $m - n$
38. If a, b, c, d are positive real numbers such that $a + b + c + d = 4$, then $M = (a + c)(b + d)$ satisfies relation
 A) $0 < M \leq 1$ B) $0 < M \leq 2$ C) $1 < M \leq 2$ D) None of these
39. If AM and GM of two positive numbers a and b are 10 and 8, respectively, find the numbers.
 A) 8, 12 B) 4, 12 C) 4, 16 D) None of these
40. The value of the expression -
 $1 - \frac{\sin^2 y}{1 + \cos y} + \frac{1 + \cos y}{\sin y} - \frac{\sin y}{1 - \cos y}$ is equal to -
 A) 0 B) 1 C) $\sin y$ D) $\cos y$

SECTION-II (Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer is in the decimals, **Mark the nearest Integer only. Have to Answer 5 questions** and questions will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

41. If $A + B = \frac{\pi}{4}$, then $(\tan A + 1)(\tan B + 1)$ equals.....
42. The interior angles of a polygon are in A.P. If the smallest angle is 120° and the common difference is 5° , then the number of sides is _____.
43. $\cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \dots + \cos 180^\circ$ is equal to k . Then the absolute value of k is...
44. If $\sec \theta = m$ and $\tan \theta = n$, then $\frac{1}{m} \left((m+n) + \frac{1}{(m+n)} \right)$ is ...
45. If the sum to first n terms of the AP 2, 4, 6, is 240, then the value of n is

Rough Work